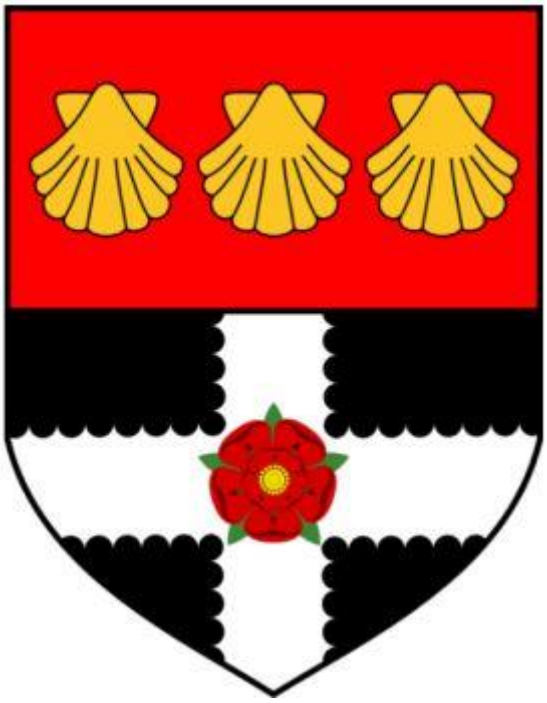


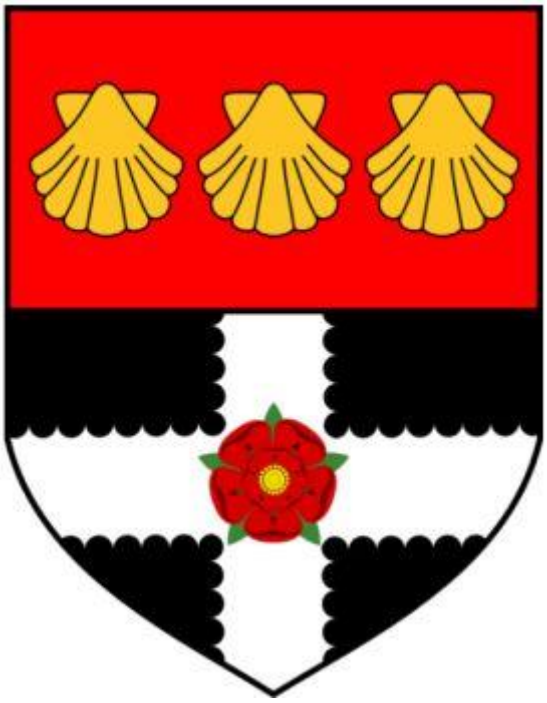
School of Biological Sciences

Project Report

MEng. Cybernetics

Investigation and Cross Comparison of methods for Independent Component Analysis

Samuel James Podmore



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Abstract

Independent Component Analysis is a prominent field of signal processing, machine learning, and system identification, producing estimates of source mixtures blind to their properties. However, in order to produce accurate estimates, the correct algorithm should be used in the correct situation to minimise error between the estimate and the reality. This process of selecting the correct method requires knowledge of both the assumptions the algorithms make on the data and how they converge to solutions. There is a vast literature in the field of ICA but little in the way of clear presentation when to use a specific algorithm. The approach of this report is to provide a clear presentation of the current developments in the field of linear non-stationary ICA such as to create a foundation for future work and allow for improved implementation of such algorithms.

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Introduction

1.1. Independent Component Analysis

Independent Component Analysis (ICA) is a blind source separation technique which attempts to create extracted signals s_i from a series of mixtures y_i of source signals $s_i$while knowing nothing about the nature of the signals. This lack of information is what in principle makes the method blind, in order create some constraint on the search space the ICA problem is often considered one of linear mixing. This linear mixing can be considered a weighting, such that the mixtures $\boldsymbol{Y}$ are the weighted sum of the sources $\boldsymbol{s}$ hence: $\boldsymbol{Y} = A\boldsymbol{s}$. Because the method is blind and uses a linear model, some assumptions must be made about the nature of the solution to ICA problem.

1.2. General Assumptions of ICA

Though each ICA algorithm will tend to assert its own assumptions on the input data, there are a few assumptions that are consistent across all blind ICA methods that assume a linear mixing model $Y = AS$.

The first is that the exact variance cannot be determined as $E\{{s_i}^2\} = 1$, which means that there are always two correct solutions, $s_i = 1$ or $s_i = -1$.Though this is not a problem in most applications it is important to note.

The second assumption can cause a little bit more trouble however. Take the model $Y = AS$, it is possible for any value taken out of the source matrix such that $Y = Q\Lambda S$ which can be rearranged as $Y = Q\tilde{S}$. All this means in practice is that the ICA algorithm will only ever be able to precisely solve the dynamics of the source, its amplitude could always be scaled. This arises because of the system being blind to the sources.

A key requirement of ICA is that the mixing matrix that is being estimated is invertible or at least pseudo-invertible. If not, then the technique will never be able to determine the mixing matrix that relates the observations to the sources. Furthermore, all sources that are being determined must be independent of one another, though this constraint is often lifted to uncorrelatedness, as it is computationally cheaper and can in cases produce better estimates.

Due to the way in which most methods of ICA deduce when to finish, random zero mean Gaussian sources are indistinguishable from white noise and therefore undetectable. Methods around this have been developed, but they often require a deviation from the independence constraint as two of the key methods for measuring independence solve till a Gaussian signal is found. See section 2.1. and 2.2. for details on kurtosis and Negentropy and how they measure gaussianity.

Due to the sheer complexity of calculating non-stationary sources and the batch nature of most ICA algorithms, a constraint that the sources must be stationary is regularly imposed, though in recursive implementations of ICA this constraint is often lifted. These assumptions and the problem of ICA can be understood intuitively through the cocktail party problem.

1.3. Cocktail Party Problem

The cocktail party problem, is considered by many an intuitive anecdote for the problems and intricacies of blind source separation [1]. In the cocktail party problem, A person is at a party and wants to listen to someone's conversation some distance away from them. Given the music is loud they'll probably hear a little of the conversation and a lot of what they consider noise. If a group of people were listening together and they all heard slightly different things, they could estimate what the person of interest was saying through combining their shared information.

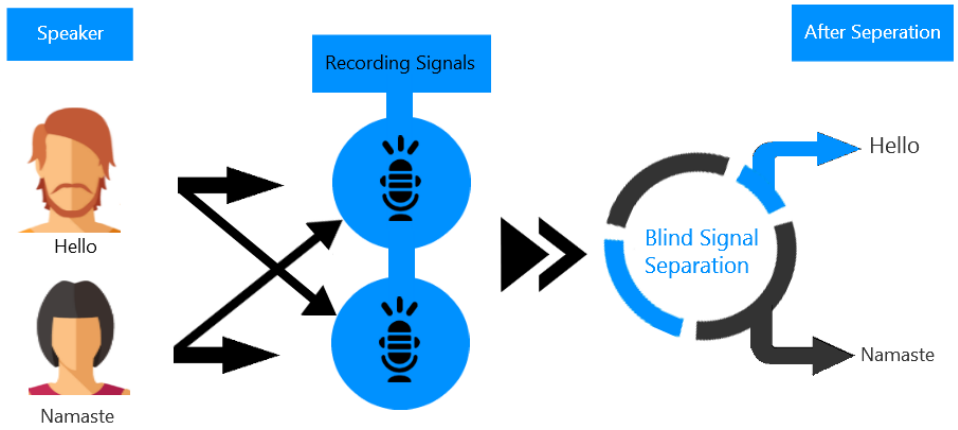


Figure 1. Illustration of the cocktail party problem [M1]

The Cocktail party problem, makes intuitive sense of the assumptions of ICA. For example, the first assumption, that the exact variances cannot be known. Without seeing the person, how would you know that they are talking to you from behind or from in front of you? Relying solely on sound you wouldn't be able to know. The second assumption, that only the dynamics can be precisely solves is also simple to understand when put in the context of the cocktail party problem. If someone is talking to you, how to do you know they are not shouting from far away, or whispering very close? Thus, the scaling Λ reflects the distance to speaker's volume ratio. The third assumption constrains those that are talking not to exist on top of one another spatially and the fourth, assumes the that people of interest are standing still as they talk.

Many consider the cocktail party problem to reflect the ICA problem and is a common method for describing the nature and intricacies of ICA and BSS in general.

Once an observation of the data Y has been made, how would you go about proving the extracted signals s_i that you've created by de-mixing A^{-1} are truly independent?

1.4. Formal Definition of Independence & Decorrelation

Two signals or probability distribution functions (pdfs) are independent if through modulation of one signal, no information about the other signal can be extracted.

$$E\left[s_1^p, s_2^q\right] = E\left[s_1^p\right]E\left[s_2^q\right] \qquad \forall q, p \in \mathbb{N}$$

It is important to note the difference between two marginal pdfs being decorrelated and independent. Statistically, decorrelation is given by two signals having a joint probability distribution $p(s_1, s_2)$ equal to zero. Hence for two signals with marginal probability distributions functions $p_{s_1}(s_1)$ and $p_{s_2}(s_2)$ respectively having a joint probability distribution of zero and hence being decorrelated is;

$$p_{s_1}(s_1)p_{s_2}(s_2) - p_{s_1 s_2}(s_1, s_2) \equiv 0$$

Decorrelated sources, such as those seen above can in fact not be independent. This is because being decorrelated is a property of being independent and independence is not a property of being decorrelated. An example for which can be seen below;

For two sources which have zero mean, zero skew and unit variance, such as $f_1(x) = x$ and $f_2(x) = x^2$, for which their relatedness is intuitive. The covariance or correlated-ness (for the relationship between these terms see appendix N.) equation for systems $f_1(x)$ and $f_2(x)$ is;

$$Cov\big(f_1(x), f_2(x)\big) = E\{f_1(x)f_2(x)\} - E\{f_1(x)\}E\{f_2(x)\}$$

$$= E\{x\}^3 - E\{x\}E\{x\}^2$$

From our definition, the skewness, $E\{x\}^3 = 0$ and the mean, $E\{x\} = 0$, $\therefore$ $Cov\big(f_1(x), f_2(x)\big) = 0$, even for two systems that are clearly not independent of one another. This is because crucially all moments of the marginal pdf's and joint pdf must be equal in order to discern whether the marginal pdfs are independent of one another. This however is computationally complex, so many algorithms relax the constraint of independence to improve computational time and potentially accuracy.

1.5. Data Whitening

Data whitening is a means of reducing the complexity of the search space by normalising the input data and zero meaning it. Data whitening consists of enforcing uncorrelatedness in the signals and limiting the variance of the data to 1. Data whitening can be computed using a simple singular value decomposition (SVD) or eigenvector technique like the one below;

Data whitening through singular value decomposition involves creating a matrix, $\tilde{x}$, such that $E\{\tilde{x}\tilde{x}^T\} = I$. The Singular Value Decomposition (SVD) method computes the covariance for a system $\tilde{x}$ such that $E\{\tilde{x}\tilde{x}^T\} = EDE^T$, $\therefore$ $\tilde{x} = ED^{-1/2}E^T x$, where E is orthogonal matrix of eigenvectors of $E\{xx^T\}$ and D is the diagonal matrix of corresponding eigenvalues.

In ICA, whitening is very important as it reduces the number of parameters the algorithms are required to estimate []. This is because the post-whitened inverting matrix A is now the orthogonal matrix $\tilde{A}$. The number of parameters required to estimate A is n^2 as the new matrix $\tilde{A}$ only requires $\frac{n(n-1)}{2}$ parameters to be estimated. After whitening, a step in which almost all ICA algorithms complete, comes the calculation of independence. However, before that it is necessary to understand the cost function paradigm.

1.6. Cost function Optimisation Paradigm

Cost function optimisation is a common concept in machine learning and more generally mathematics, focusing on the minimisation or maximisation of a cost function of interest. The method of this optimisation might be a simple gradient decent method or in more complex unknown cases a stochastic method such as Least Mean Squares. Cost function optimisation problems can be formulated explicitly and implicitly depending on the nature of the problem and derivation but generally come in the form;

$$X(n+1) = Cost(X(n)) + Optimisation(X(n))$$

There is an intuitive relationship between this structure and typical numerical analysis methods such as Runge-Kutta or forward Euler:

$$X(n+1) = X(n) + \Delta h \frac{\partial x}{\partial n}$$

Where the optimisation is complete at the point at which $\frac{\partial x}{\partial n} = 0$ and $X(n+1) = X(n)$. In some cases, more complex solutions rather than finding minima or maxima are required, however they will not be included in this report as they are out of the scale of it.

The analysis of the algorithms is structured as to mimic the cost optimisation paradigm, first exploring the various cost functions related to independence and then exploring the optimisation methods employed.

Costs of Independence

2.1. Moments

Moments are a quantitative measure of the distribution of a set of points. Different moments describe different properties of the distribution. The Zero-th moment is always one, as it described the absolute probability of the set. The first moment, is the mean, the second is variance, the third is skewness and the fourth is kurtosis. There are names associated to higher order moments but they will not be discussed as they are not relevant. Below is the equation for the nth generalised standard moment;

$$M_n = \frac{E\{(x-\mu)^n\}}{\sigma^n}$$

This standardised moment equation is used to calculate all higher order moments, as they all follow the general rule described above. Where x is a probability distribution, μ is the mean of x, σ

is the standard deviation of x and $E[\cdot]$ denotes the expectation operator. To calculate using this general standardised moment equation one must know the Raw moment, the mean, μ via;

$$\mu(x) = \frac{\sum x_i}{n} \equiv E[x]$$

With the mean, it becomes possible to calculate the variance of the probability distribution using the central moment equation;

$$\sigma(x) = \frac{\sum(x_i - \mu)^2}{n} \equiv E[(x-\mu)^2]$$

From these two equations, some very important measures of a probability distribution can be calculated. The third order standardised moment, the skewness describes the proportion of the distribution that are skewed to any one side. If skewness, $\gamma > 0$, the distribution is skewed towards smaller values. If $\gamma < 0$, then the distribution is skewed towards larger values. The equation for skewness can be seen below;

$$\gamma(x) = E[(\frac{x-\mu}{\sigma})^3]$$

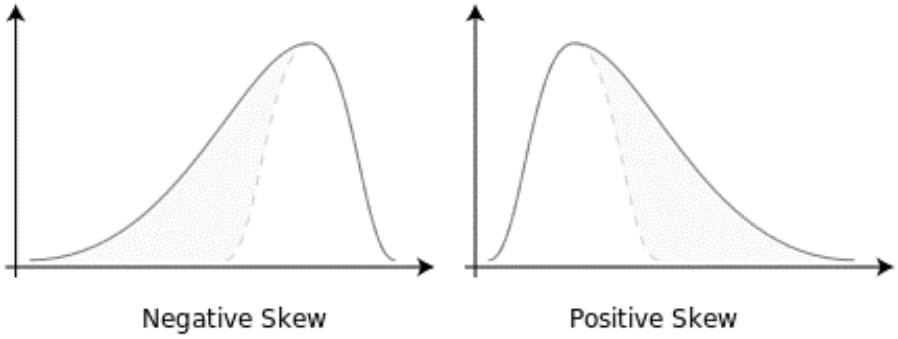


Figure 2. is a graphical representation of skewness [M2]

The Fourth Order standardised moment is kurtosis, the quantitative measure of how "spiky" a distribution is. If $\mathrm{K}(x) > 0$, the associated probability distribution, x, will have a spiky distribution like that seen in Figure N.1. If $\mathrm{K}(x) < 0$, the distribution will be flatter and wider than a Gaussian distribution as in Figure N.2.

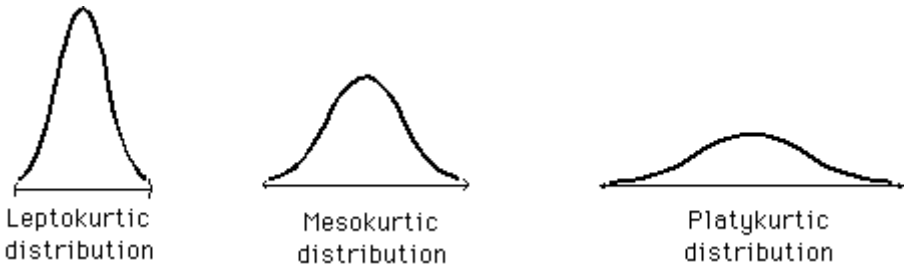


Figure 3. graphical representation of pdfs of different values of kurtosis [M3]

Kurtosis, and the modified excess kurtosis (which is simply kurtosis minus three) is widely used in band-pass filters, image analysis and ICA. The equation for kurtosis can be found below;

$$\mathrm{K}(x) = E[(\frac{x-\mu}{\sigma})^4]$$

This measure of non-gaussianity is used as a measure of independence through some simple logic related to the central limit theorem. One of the main assumptions when using kurtosis as a cost for independence is to note that, per the central limit theorem any mixture of sources will tend to be more Gaussian and complex than the source signals that constitute it. Thus, as kurtosis is a measure of gaussianity, if a value has a very large magnitude of kurtosis, it must be independent as it is as least Gaussian as it can possibly be. However, kurtosis is non-robust in nature, which means it is sensitive to outliers in the input data set. Entropy, a more robust measure of nongaussianity is also commonly used in the field of ICA, though at the cost of computational complexity.

2.1.1. Cumulants

Cumulants are an extension of moments that consider the spread of the data and are easily calculated from the moment equation, as below;

$$C_1 = M_1$$

$$C_2 = M_2 - M_1^2$$

$$C_3 = 2M_1^3 - 3M_1M_2 + M_3$$

$$C_4 = -6M_1^4 + 12M_1^2M_2 - 3M_2^2 - 4M_1M_3 + M_4$$

In the case of a signal x which is zero mean and of unit variance, the fourth cumulant is the exact kurtosis;

$$C_4(x) = M_4(x) - 3 \equiv K(x) - 3$$

2.2. Entropy

Information entropy is the average measure of surprise [] across a probability distribution function (pdf) and is a measure of the uniformity of a pdf.

$$H(\boldsymbol{X}) = \frac{1}{N}\sum_{i}^{N} p(x_i)\log\left[\frac{1}{p(x_i)}\right]$$

Taking a similar assumption about the central limit theorem to kurtosis, entropy measures the gaussianity of a system as entropy is maximised when a distribution is Gaussian []. The concept of entropy is easy to understand when given an example. Take for example the entropy of a fair coin who has the probability distribution function P_{fc} as calculated by the equation for entropy;

$$H(p_{fc}) = \frac{1}{N}\sum P_{fc}\log(P_{fc}) = (0.5)\log\left[\frac{1}{(0.5)}\right] + (0.5)\log\left[\frac{1}{(0.5)}\right] = 1\ bit\ per\ coin\ flip$$

However, if we look at the entropy of an unfair coin who has a pdf P_{bc} who has a probability of heads being 0.9 then;

$$H(P_{bc}) = (0.9)\log\left[\frac{1}{0.9}\right] + (0.1)\log\left[\frac{1}{0.1}\right] = 0.469\ bits\ per\ coin\ flip$$

This decrease in bits per coin flip is due to the redistribution of probabilities in the pdf of the coin. This is related to the median of the coin in this example, if the coin is flipped, the expected result is heads, due to the skewed probability, thus the result of tails is more surprising.

Entropy for a joint probability distribution $P(\boldsymbol{X}, \boldsymbol{Y})$ is a generalised version of the entropy of a single variable;

$$H(\boldsymbol{X}, \boldsymbol{Y}) = E\left[\log\left(\frac{1}{p(x,y)}\right)\right]$$

If the sources $\boldsymbol{X}$ and $\boldsymbol{Y}$ are independent, then;

$$H(\boldsymbol{X}) + H(\boldsymbol{Y}) = \int_x \int_y p(x)p(y)\log\left(\frac{1}{p(x)p(y)}\right) dydx = E\left[\log\left(\frac{1}{p(x)}\right)\right] + E\left[\log\left(\frac{1}{p(y)}\right)\right]$$

Entropy is intrinsically linked to mutual information of two systems. The mutual information between two variables $\boldsymbol{X}$ and $\boldsymbol{Y}$ is the average reduction in uncertainty about the value of $\boldsymbol{X}$ provided by a value of $\boldsymbol{Y}$ and *vice versa*.

$$I(\boldsymbol{X}, \boldsymbol{Y}) = H(\boldsymbol{Y}) + H(\boldsymbol{X}) - H(\boldsymbol{X}, \boldsymbol{Y})$$

$$I(\boldsymbol{X}, \boldsymbol{Y}) = E\left[\log\left(\frac{1}{p(x)}\right)\right] + E\left[\log\left(\frac{1}{p(y)}\right)\right] - E\left[\log\left(\frac{1}{p(x)p(y)}\right)\right]$$

Whilst mutual information measures the shared information between two probability distributions there is another standard measure of the difference between two probability distributions, called the relative entropy or Kullback-Leibler Divergence.

2.2.1. Kullback-Leibler Divergence

The Kullback-Leibler divergence is a measure of the distance between an extracted signal y by a given unmixing matrix W and another, typically Gaussian, distribution. The Aim when using the Kullback-Leibler divergence is to find a matrix W such that the relative entropy is at its minimum value $H(Y)$ and hence ensure that $P_y(y^t) \approx P_s(y^t)$. Below is the equation of the relative entropy between distribution P_y and the model distribution P_s;

$$H(Y) = -\frac{1}{N}\sum_t^N \log\left(\frac{P_y(y^t)}{P_s(y^t)}\right) \equiv -E\left[\log\left(\frac{P_y(y^t)}{P_s(y^t)}\right)\right]$$

Entropy can be used to find independent components in several ways. A strategy for finding independence of a set of mixtures, commonly referred to as the infomax method finds an unmixing weight matrix W so as to maximise the entropy (of a non-linear function $g(\cdot)$) of the signals extracted by said unmixing matrix. The unmixing matrix that maximises the entropy is also maximising the mutual information between the extracted signals and the signal mixtures.

Optimisation Methods

Optimisation methods attempt to find the minima or maxima of a search space by finding the point in which $\frac{dy}{dx} = 0$. There are two main categories, stochastic and deterministic. Stochastic methods cannot guarantee convergence to the point at which $\frac{dy}{dx} = 0$ but take no prior knowledge

about the distribution of the search space. Deterministic methods have guaranteed convergence at the cost of requiring knowledge of the search space. Because of this required information they are rarely implemented.

3.1. Brute Force Exhaustive Search

The brute force exhaustive search method is an incredibly inefficient one, exhausting the search space by cycling through all possible values. Though convergence is guaranteed, it may not be feasible as the computational power required/time required may be very large. Because of this the brute force exhaustive search method is never used, cleverer techniques are more often employed, exploiting the nature of the search space using assumptions and statistics.

3.2. Steepest Decent

Steepest decent is an iterative deterministic method of optimisation in linear filtering finding a weight matrix W such that $y = Wx$. The method requires that the statistics of the mapping be known, such as the cross correlation and autocorrelation. Because of this it is not suitable for blind source separation but it is an important method none the less. When using the mean squared error (MSE) measure of distance between y and x, the gradient vector ∇J that reduces the MSE the most is the direction of steepest decent. Hence;

$$J(w) = E\{[d(n) - w^H u(n)][d(n) - w^H u(n)]^*\}$$

Where;

$R = E\{u(n)u^H(n)\}, p = E\{u(n)d^*(n)\}$ and $\sigma_d^2 = E\{d(n)d^*(n)\}$ then;

$$J(w) = \sigma_d^2 - w^H p - p^H w + w^H R w$$

As $\nabla J(w) = 2\frac{dJ(w)}{dw^*} = -2p + 2Rw$. Here p denotes the exact cross correlation and R the exact autocorrelation. Thus, in the general cost optimisation formula;

$$w(n+1) = w(n) + \frac{1}{2}\mu(-\nabla \mathrm{J(w)})$$

Through consecutive iterations, eventually $\nabla J(w) \to 0$ and $w(n+1) = w(n)$

3.3. Least Mean Squares (LMS)

Least Mean squares is a natural extension to the method of steepest decent, using approximate autocorrelations and cross correlations rather than exacts ones. This is important when using real data as often the statistics are unknown. Because of this lack of information, the least mean squares method is a stochastic iterative method of gradient decent rather than a deterministic one, meaning convergence is not guaranteed. Using the estimates at each time step, the direction of steepest decent can be approximated as;

$$\hat{R} = E\{u(n)u^*(n)\} \text{ and } \hat{p} = E\{u(n)d^*(n)\}$$

$$\hat{\nabla}J(n) = -2\hat{p}(n) - 2\hat{R}(n)\hat{w}(n)$$

Hence;

$$\hat{w}(n+1) = \hat{w}(n) + \frac{1}{2}\mu(-\hat{\nabla}J(n))$$

Because of this approximation, the convergence of the LMS algorithm is not always guaranteed, and often, the LMS fluctuates around a series of possible solutions. LMS is often implemented recursively, in which case it's called recursive least squares (RLS).

3.4. Recursive Least Squares

Recursive Least Squares (RLS) is a stochastic gradient descent method for finding local minima or maxima through recursively updating a combination of a previous value and a weight cross-correlation. It takes the general form of a linear filter relating finding values of w_i in order to relate the target y_i to input u_i. This is written mathematically as;

$$y_i = \sum w_i u_{k-i}$$

The method measures error between the desired state and the current state as the sum of squared errors and uses a learning rate, $\beta(n.i) = \lambda_i^n$ where $0 < \lambda < 1$ to allow for adaptability. Hence;

$$\varepsilon(n) = \sum_{i=i_1}^{n} \beta(n,i)[d(i) - \sum_{k=o}^{M-1} w_k u(i-k)]^2$$

This model is then fit to the least squares general form for $Y = AS$, $A = [SS^T\}^{-1}S^TY$. Our current model when adapted to this form can be shown as;

$$u'(i) = \sqrt{\lambda^{n-i}}u(i) \quad \text{and} \quad d'(i) = \sqrt{\lambda^{n-i}}d(i)$$

$$w(n) = (\sum_{i=i_1}^{n} u'(i)u'(i)^T)^{-1} \sum_{i=i_1}^{n} u'(i)d'(i) \equiv [\phi(n)]^{-1}\psi(n)$$

Here $\phi(n)$ is the autocorrelation of the inputs and $\psi(n)$ is the cross-correlation of the desired and current values. This model for $w(n)$ is then written into a solely recursive form.

$$w(n) = [\phi(n)]^{-1}\psi(n) \quad \therefore \quad w(n-1) = [\phi(n-1)]^{-1}\psi(n-1)$$

Now rewrite $\phi(n) \in \phi(n-1)$ and $\psi(n) \in \psi(n-1)$

$$\phi(n) = \lambda \sum_{i=1}^{n-1} \lambda^{n-1-i}u(i)u(i)^T + u(n)u(n)^T = \lambda\phi(n-1) + u(n)u(n)^T$$

$$\psi(n) = \lambda \sum_{i=1}^{n-1} \lambda^{n-1-i}u(i)d(i)^T + u(n)d(n)^T = \lambda\psi(n-1) + u(n)d(n)^T$$

Now through application of the law of matrix inversion, the computational time of calculating $[\phi(n)]^{-1}$ can be reduced significantly. Hence;

$$\phi(n)^{-1} = \lambda^{-1}\phi^{-1}(n-1) - \frac{\lambda^{-2}\phi^{-1}(n-1)u(n)u(n)^T\phi^{-1}(n-1)}{1 + \lambda^{-1}u^T(n)\phi^{-1}(n-1)u(n)} \equiv P(n)$$

By creating a new variable $k(n)$ we can create a recursive version of $P(n)$, hence;

$$k(n) = \frac{\lambda^{-1}P(n-1)u(n)}{1 + \lambda^{-1}u(n)^TP(n-1)u(n)} \equiv P(n)u(n)$$

$$P(n) = \lambda^{-1}P(n-1) - \lambda^{-1}k(n)u(n)^TP(n-1)$$

Now through a series of simple algebra the creation of a recursive update rule for $w(n)$ based on previous values $w(n-1)$ can be formed;

$$\begin{aligned} w(n) &= [\phi(n)]^{-1}\psi(n) = P(n)\big(\lambda\psi(n-1) + u(n)d(n)\big) \\ &= P(n)\big(\lambda\phi(n-1)w(n-1) + u(n)d(n)\big) \\ &= P(n)((\phi(n) - u(n)u(n)^T)w(n-1) + u(n)d(n)) \\ &= w(n-1) - P(n)u(n)u(n)^Tw(n-1) + P(n)u(n)d(n) \\ &= w(n-1) + P(n)u(n)(d(n) - u(n)^Tw(n-1)) \\ &= w(n-1) + P(n)u(n)\alpha(n) \\ &= w(n-1) + k(n)\alpha(n) \end{aligned}$$

Where $a(n) = d(n) - u(n)^Tw(n-1)$

This is the complete derivation of the recursive least squares algorithm for gradient descent calculations. This general form has been modified many times [2], [3] and has seen no end of application in a wide variety of fields [4].

ICA Through Optimisation

ICA comes in many forms and flavours, but to convey a clear understanding of the field they will be organised loosely by the methods to which they optimise their cost functions. The first discussed algorithm will be the Herault-Jutten method due to both its simplicity and historical relevance.

4.1. Herault-Jutten Method

The Herault-Jutten (HJ) method [5] [6] of ICA was the first developed, following from the then recent developments in adaptive filtering and neural networks it minimised the linear cost function $y(t) = x(t) - \widehat{W}(t)y(t)$., which through rearrangement is $y(t) = [I + \widehat{W}(t)]^{-1}x(t)$. To those familiar with control diagrams, the algorithm can be understood simply from below;

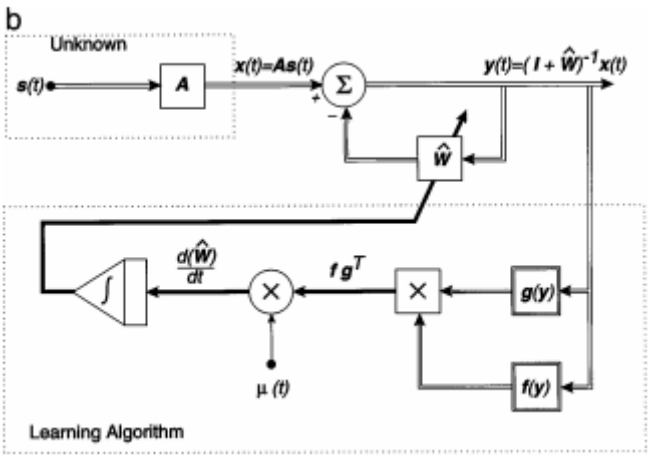


Figure 4. Schematic of the H-J method [6]

On the basis of their cost criteria, they developed an adaptive learning algorithm;

$$\frac{dW}{dt} = \mu(t) f[y(t)] g^T[y(t)]$$

Where $F[\cdot]$ and $g[\cdot]$ are the non-linear odd activation functions y^3 and $\tanh(10y)$ respectively and the learning rate $\mu(t) > 0$.The "independence" is measured through minimisation of the covariance matrix of the non-linear odd functions $f(y_i)$ and $g(y_j)$. However, if the sources have even probability distribution then $E[f(y_i)]E[g(y_j)] \equiv 0$ and hence no approximation of the decorrelated components can be found. This in part is due to the fact that the method doesn't use a true definition of statistical independence. If the functions $f(y_i)$ and $g(y_j)$ are chosen to both be y then the H-J method would be completing a PCA on the given data. For more information on PCA see appendix 4. The Herault-Jutten algorithm is typically implemented using the forward Euler numerical method, hence;

$$W_{n+1} = W_n + \Delta t \frac{dW}{dt}$$

Where t is some positive constant chosen carefully as to maximise convergence time and reduce oscillations.

As the Herault-Jutten method measures some pseudo-measure of independence and is notably inefficient therefore it is rarely used. In order to produce good approximations of the source signals many method use higher order statistics, in an attempt to maximise the mutual information between the mixture and independent component.

4.2. Infomax method

The Infomax method [7] maximises the mutual information between the sources and the mixture distribution whilst insuring that the sources are as independent as possible with one another. For a given observed signal, $y = Wx$, find W such as to maximise the entropy variable $Y = g(y)$ where $g(.)$ Is typically some non-linear function. For a Finite set of samples from the distribution with pdf P_Y entropy can be estimated as;

$$H(\boldsymbol{Y}) = -\frac{1}{N}\sum_{t}^{N}\ln\left(P_Y(\boldsymbol{Y}^t)\right) \equiv -\frac{1}{N}\sum_{t}^{N}\ln\left(\frac{P_y(y^t)}{P_s(y^t)}\right)$$

Hence if an optimal unmixing matrix W exists then $P_y(y^t) = P_s(y^t)$. This implies that the joint pdf $P_Y(Y)$ is as uniform and that its entropy is maximised by the optimal matrix W meaning that the variables Y are independent. This in turn implies that $y = g^{-1}(Y)$ are also independent.

Thus P_s acts as a model function, as the W that maximises the entropy is the W that creates sources y whose pdf satisfies $P_y = P_s$. Substituting $P_y(y) = \frac{P_x(x)}{|W|}$ into the equation for entropy above;

$$H(\boldsymbol{Y}) = -\frac{1}{N}\sum_{t}^{N}\ln\left(\frac{P_x(x^t)}{|W|P_s(y^t)}\right) \equiv \frac{1}{N}\sum_{t}^{N}\ln\left(P_s(y^t)\right) + \ln(|\boldsymbol{W}|) - \frac{1}{N}\sum_{t}^{N}\ln\left(P_x(x^t)\right)$$

$$H(Y) = \frac{1}{N}\sum_{t}^{N}\ln\left(P_s(y^t)\right) + \ln(|W|) + H(x)$$

$$H(\boldsymbol{Y}) - H(x) = h(\boldsymbol{Y}) \equiv \frac{1}{N}\sum_{t}^{N}\ln\left(P_s(y^t)\right) + \ln|\boldsymbol{W}|$$

If the M marginal pdfs of the model joint pdf P_s are assumed independent and we substitute the commonly used super-Gaussian (High kurtosis) model pdf $P_s = (1 - \tanh(s)^2)$ then;

$$h(\boldsymbol{Y}) = \frac{1}{N}\sum_{i}^{M}\sum_{t}^{N}\ln(1 - \tanh\left(w_i^T x^t\right)^2) + \ln(|\boldsymbol{W}|)$$

Thus, the solution for $h(Y)$ exists at the optimal weighting W such that it minimises the Kullback-Leibler divergence between the pdf of the observed variable P_y and the model's pdf P_s. Another method, typically implemented using entropy is the Log-Likelihood method.

4.3. Loglikelihood method

The loglikelihood method [8] relies crucially on the loglikelihood equation which is a measure of the shared information between two systems:

$$\frac{1}{N}\ln\left(L(\boldsymbol{W})\right) = \frac{1}{N}\sum_{i}^{M}\sum_{t}^{N}\ln\left(P_s\left(w_i^T x^t\right)\right) + \ln(|\boldsymbol{W}|)$$

For a given observed signal $x = As$, find W such that the joint pdf, $P_x(x)$ is as similar as possible to the joint pdf P_s of the unknown source signals s. Hence;

$$x = As \;\&\; s = Wx$$

$$P_x(x) = P_s(s)\left|\frac{\partial s}{\partial x}\right| \equiv P_s(\boldsymbol{W}x)|\boldsymbol{W}| \;\therefore \max\left(L(\boldsymbol{W})\right) \rightarrow s_1, \cdots, s_n \; are \; independent$$

Given that P_s is the product of its M marginal pdfs and assuming the sources are independent. If we substitute the commonly used super-Gaussian model $P_s = (1 - \tanh(w_i^T x^t)^2)$

$$\frac{1}{N}\ln\big(L(\boldsymbol{W})\big) = \frac{1}{N}\sum_{i}^{M}\sum_{t}^{N}\ln\big(1-\tanh(w_i^T x^t)^2\big) + \ln(|\boldsymbol{W}|)$$

Which is identical to the Infomax model. This relationship between the loglikelihood method [] and infomax method [] was not discovered until much after release of both methods [].

The minimisation of mutual information is considered computationally complex as estimates of the pdfs of the outputs from a finite set of data are required. The main approaches to overcome this involve the truncation of Edgeworth or Gram-Charlier [9] expansions of the pdf. These approaches however don't produce good estimates and because of this, methods such as the INFOMAX and mutual information can fail to separate signals [10].

4.4. FASTICA method

FastICA [11] is a very common method for ICA because of its good convergence time [12], [13] its rich literature regarding its modifiability [14], [15] and sequential/deflation approach to source extraction. The FastICA method is closely related to Projection Pursuit [16], [17] and is concerned with finding a direction in the data which maximises its measure of nongaussianity. The method uses artificial neurons [18] with weights w_i to extract the sources by modifying these weightings with a learning based approximate Newton iterations.

FastICA measures nongaussianity with a modified variant of entropy called negentropy. Negentropy is the difference in entropy between a Gaussian distribution (which has maximum entropy) and the extracted signal. Hence;

$$J(\boldsymbol{Y}) = H(\boldsymbol{Y}_{Gauss}) - H(\boldsymbol{Y})$$

However, this measure is difficult to compute [1] and which lead to the approximate method seen below;

$$J(\boldsymbol{Y}) \approx \frac{1}{12}E[\boldsymbol{Y}^3]^2 + \frac{1}{48}kurt(\boldsymbol{Y})^2$$

This equation however features Kurtosis, which as described earlier performs badly in the presence of outliers. In [19] a new approximate was formulated based on the principles of maximum entropy;

$$J(y) \approx \sum_{i+1}^{p} k\{E[G(y)] - E[G(y_{gauss})]\}^2$$

Where k is some positive constant, the variable y is whitened and hence zero mean and unit variance and $G(\cdot)$ are some non-quadratic functions [1]. Even though this approximation is not very accurate with a well-chosen function G is can be consistent with other measures of nongaussianity in the sense that it is always positive and only zero if its inputs is Gaussian. Two examples of typical functions for G are $G_1(\cdot)$ and $G_2(\cdot)$ as seen below;

$$G_1(u) = \frac{1}{\alpha_1}\log(cosh(a_1 u)) \qquad G_2(u) = e^{\frac{-u^2}{2}} \qquad 1 \le \alpha \le 2$$

The derivatives g_i of these model functions G_i are;

$$g_1(u) = \tanh(a_1 u) \qquad g_2(u) = ue^{\frac{-u^2}{2}} \qquad 1 \leq a \leq 2$$

The FastICA process for a single source is to find the neuron weighting w such that the extracted source has maximized negentropy as measured by $J(y)$. The learning process is subject to a single constraint that $|w^T x|$ = 1 which if the data has been whitened is the same as the weight matrix having a norm of unity. FastICA for a single source has its weights initialised randomly then follows a 3-step process:

1) $w^+ = E[xg(w^T x)] - E[g'(w^T x)]w$
2) $w = \frac{w^+}{||w^+||}$
3) Repeat from 1) if w has not converged.

However, this process must be adjusted for multiple sources to stop the neurons converging to the same source. The FastICA employs an extra decorrelation step after every iteration to ensure that the weights find unique solutions. There are many methods achieving this and the one most commonly employed is a deflation scheme based on Gram-Schmidt correlation. This deflation means that the sources are extracted one by one. When p weight vectors $w_1, \cdots, w_p$ have been found, FastICA begins finding the value of the next weighting w_{p+1} and after every iteration subtracts from w_{p+1} the projections of $w_{p+1}^T w_j w_j$ of the previously estimated p weights and then renormalizes w_{p+1}, hence:

$$w_{p+1} = w_{p+1} - \sum_{j=1}^{p} w_{p+1}^T w_j w_j$$

$$w_{p+1} = \frac{w_{p+1}}{\sqrt{w_{P+!}^T w_{P+1}}}$$

As FastICA solves for an approximate value of negentropy it doesn't truly solve for independence, but rather something similar to it. In cases where strict independence is required through optimisation without using entropy, the ERICA method can prove quite useful.

4.5. ERICA (Equivalent robust ICA) Method

ERICA [10], [20] is in principle an extension of the infomax method using a modified criterion for independence. Because the marginal entropies of the outputs $h(y_i)$ are difficult to estimate, ERICA replaces each differential entropy in the Infomax cost function by its respective $(1+\beta)$-order cumulant of the outputs, Hence;

$$h(y_i) \to \frac{|C_{y_i}^{1+\beta}|}{1+\beta}$$

By applying this cumulant measure of independence to the mutual information explored previously the cost function $\psi(W)$ is created:

$$\psi(W) = \sum_{i=1}^{N} \frac{|C_{y_i}^{1+\beta}|}{1+\beta} - \log|det(W)| - h(x)$$

The convergence for the source separating function $\psi(W)$ is a saddle point and because of this, ERICA uses Quasi-Newton methods of optimisation rather than the gradient decent methods employed by other algorithms. ERICA also doesn't solve for an exactly orthogonal de-mixing matrix as it is not computational cheap in the presence of additive and potentially spatially correlated Gaussian noise.

Entropy enforces strong independence on the extracted sources and hence solves for independent components rather than decorrelated ones. Unlike kurtosis, Entropy and its variants are robust and don't suffer from poor source estimation in the presence of outliers however due to inaccuracies in the approximations of the expectation operators poor components can be produced when working with entropy. Because in practical settings the expectation operator takes a sample mean rather than a true mean, entropy derived methods require large data sets to find good approximations of the source signals hence don't lend themselves to real time implementations but rather batch ones.

ICA through Joint Diagonalization

5.1. Joint Diagonalization

Joint Diagonalization is the name given to the process of simultaneously zero-ing off the non-diagonal components of a series of matrices. Many methods of ICA use Joint Diagonalization as it is computationally effective, especially on whitened data. Though there are many methods of joint diagonalization [21] [22] [23], this report will only explore one, the givens rotation approach to joint diagonalization.

5.2. Givens Rotations

Givens Rotations are a method for reducing the off-diagonal components of a target matrix to zero through a repeated series of rotational matrices as;

$$\begin{bmatrix} cos\theta & sin\theta \\ -sin\theta & cos\theta \end{bmatrix} = 1$$

For example, say the desired rotation would reduce component y_2 of the 2×1 matrix y to zero.

$$\begin{bmatrix} y_1 \\ 0 \end{bmatrix} = \begin{bmatrix} cos\theta & sin\theta \\ -sin\theta & cos\theta \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$\therefore$

$$y_1 = x_1 cos\theta + x_2 sin\theta \text{ and } x_2 cos\theta - x_1 sin\theta = 0 \rightarrow \theta = \arctan(\frac{x_2}{x_1})$$

This can be done on a much larger matrix through careful placement of the $sine$ and $cosine$ components. For example, to remove a value from a 3×1 matrix, the rotational matrix would be;

$$\begin{bmatrix} y_1 \\ y_2 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & cos\theta & sin\theta \\ 0 & -sin\theta & cos\vartheta \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

This rotation can be done to any matrix, and is often done repeatedly, until all that is left is a diagonal matrix.

Givens rotations have an intuitive link to the projection pursuit or Gram-Schmidt orthogonalization methods described earlier. Many ICA methods implement JD with varying degrees of complexity. An extremely simple method would be the AMUSE method.

5.3. Amuse Method

The Amuse method [24] is a computationally cheap one that uses second order statistics to enforce decorrelation across a snapshot of its input y. Although this method produces poor approximations it is very robust, as it is only required to find a transformation matrix between two covariance matrices. The Amuse method exploits spatio-temporal decorrelation through second order statistics, employing a cascade SVD to ensure decorrelation of two time-lagged covariance matrices. The process is described below;

1) First SVD is employed to whiten the observed data $x(t)$, through a linear mapping Q such that $x_1(t) = Qx(t)$. Hence;
$$Q = R_{xx}(k)^{-1/2} = E[x(k)x^T(k)]^{-1/2} = (V\Lambda V^T)^{-1/2} = V\Lambda^{-1/2}V^T$$
2) Amuse then employs SVD a second time on a time lagged covariance matrix $R_{xx}(k,\tau)$;
$$R_{xx}(k,\tau) = E[x(k)x^T(k-\tau)] = U\Sigma V^T$$
3) From which a unmixing matrix W can be generated such that $W = U^T Q$.

The simplicity of this method both an advantage and a disadvantage. The small percentage of the data being tested means that it is computational cheap but susceptible to outliers in the covariance matrix. Furthermore, the AMUSE method produces poor approximations of the dynamics for each component because of its reliance on a small sample of the data. By increasing the amount covariance matrices being joint diagonalized it's possible to increase the accuracy of the approximations at the cost of computational time. This is the methodology of the TDSEP approach to blind source separation.

5.4. TDSEP Method

The TDSEP (Temporal Decorrelation SEParation) [25] method can be seen in many ways as an extension of the Amuse method and is closely related to the H-J method. TDSEP aims to produce a series of decorrelated signals through joint diagonalization of several time lagged correlation matrices. Because this method maximises the decorrelation across a series of correlation matrices it doesn't solve for independence. Hence;

$$l(C_{ij}) = \sum_{i \neq j} \mathrm{E}[y_i(t)y_j(t)]^2 + \sum_{k=1}^{K} \sum_{i \neq j} E[y_i(t)y_j(t+\tau_k)]^2$$

When using TDSEP It is important to select lags that contain maximally different information such as to capture most of the information in the input data set. Though TDSEP doesn't solve for any

measure of independence it produces very good approximations of the independent components in cases where there is a temporal relationship in the data. Another, very commonly used decorrelation method that is very similar to TDSEP is SOBI.

5.5. SOBI Method

The SOBI (Second Order Blind identification) method [26] attempts to joint diagonalized several time lagged covariance matrices such as to ensure a maximum decorrelation across its components. The methodology of SOBI comes as a natural extension to that seen in TDSEP. For a set of observed signals $y(t)$ which are constituted of linear mixing A of source signals x, then $y(t) = Ax(t) + n$ where n is zero mean spatially white Gaussian noise. The covariance matrices R at various lags τ of the observed signals $y(t)$ are as follows;

$$R(0) = AR_s(0)A^H$$

$$R(\tau) = AR_s(\tau)A^H$$

Therefore, a post whitened observed set $y(t)$ whitened by matrix W produces;

$$E[Wy(t)y^T(t)W^T] = WAA^TW^T = I \ \therefore A = \ W^{\#}U$$

Where # denotes the Moore-Penrose pseudo inverse. This spatial whitening produces covariance matrices $\underline{R}(\tau)$ for $\forall\tau \neq 0$ such that; $\underline{R}(\tau) = UR_s(\tau)U^H$. If we define the off diagonal components of a $n \times n$ matrix M as, $off(M) = \sum_{1\leq i\neq j\leq n} |M_{ij}|^2$ then joint diagonalization of several time lagged covariance matrices $\underline{R}(\tau)$ by a unitary matrix V is;

$$C(\underline{R}, V) = \sum_{\tau} off(V^H\underline{R}(\tau)V)$$

In practice the covariance matrices $\underline{R}$ need not be joint diagonalizable as a set or individually. The difference between SOBI and TDSEP is subtle. Where TDSEP looks at both the current covariance and a series of time lagged ones SOBI looks only at all the time lagged covariance matrices.

Where kurtosis would provide poor approximations of the independent component it may be possible to improve the approximation by using the 4th order cumulant rather than the 4th order moment. This cumulant takes into account the spread of the data and is the principle behind the JADE method.

5.6. JADE Method

Jade [27] is a popular method for ICA, as it enforces a strict measure of independence through using fourth order cumulants whilst remaining computationally cheap by only joint diagonalizing a small sub-set of the available cumulant matrices. First the data is whitened such that $z(t) = Us(t) + Wn(t)$. Then the data is passed to the fourth order cumulant function Q_z as described below;

$$Q_z(M) = E[z^*Mzzz^*] - \ R_zMR_z - \ R_zTrace(MR_z)$$

In noiseless cases, $R_z = UR_sU^H = UU^H = I$, such that $Q_z(I_n) = E[|z|^2zz^*] - (n+1)I_n$. From this, it is clear that $Q_z(M)$ and the weighted covariance must have equal eigenvectors. Hence

U may be identified as a unitary diagonaliser without computing the full cumulant matrix. In the noisy case jade is implemented in a series of steps. First Jade whitens the observed data $Y(t)$ through SVD creating the post whitened matrix $z(t) = \widehat{W}Y = Us(t) + \widehat{W}(t)$. Then computes a series of sample fourth order cumulants L_z from the whitened data $z(t)$. Jade then joint diagonalises the set of sample cumulants by a unitary matrix $\widehat{U}$. Finally, Jade constructs a de-mixing matrix $\hat{A} = \widehat{W}^{\#}\widehat{U}$ where $^{\#}$ denotes the Moore-Penrose pseudoinverse.

Why and When?

There are many fields in which ICA can be practically applied, but as has been explored all ICA algorithms make assumptions on the nature of the dataset and have different methods for solving the BSS problem. If the dataset has a temporal correlation then TDSEP which assumes a temporal correlation should perform well. However, methods that assume strict independence across the components, such as EFICA should perform poorly. Hence it is important to both understand the algorithms and the dataset that is undergoing source separation.

Literature Review

There is a great deal of literature in cross comparisons of ICA algorithms however there is no standardisation amongst the metrics used to define success for any quality of the algorithms. With widely different metrics of success and landscapes in which these algorithms are being applied it becomes very difficult to grasp a clear understanding of the techniques and how they relate to one another both in structure and application.

In systems without noise, the optimal solutions exist as an unmixing matrix equal to the inverse of the mixing matrix. This however, is not the case in a system with noise, instead, the optimality of the solution depends on the criteria by which the distributions sources are estimated [28]**.** Because of this, many authors measure quality of independent components as the sum of squared errors of either the estimated and actual unmixing matrices or of the estimated and actual component distributions. Koldovsky et al [28] propose a minimum SSE approach to measuring success via calculating error in the estimated mixing matrix W:

$$SINR_k = \frac{(WA)_{kk}^2}{\sum_{i\neq k}^{d}(WA)_{ki}^2 + \sigma^2 \sum_{i=1}^{d} W_{ki}^2}$$

Which at $W^{MMSE} = A^T(AA^T + \sigma^2 I)^{-1}$

$$min\ SINR_k = \frac{G_{kk}^2}{\sum_{i\neq k}^{d} G_{ki}^2 + \sigma^2 \sum_{i=1}^{d}(GA^{-1})_{ki}^2}$$

Where $G = (I + \sigma^2(A^TA)^{-1})^{-1}$. Koldovsky et al implement this measure of success testing 11 algorithms on four nonstationary acoustic data mixtures [29]. In this case, BGL [30] produced the most estimate component with the least error, closely followed by BARBI [31] and then EFICA. It is important to note what each algorithm was built for when testing it, as EFICA wasn't designed to deal with nonstationary sources unlike BGL and BARBI but did score very well with this Signal Interference Noise Ratio measure of success.

A Delorme, creator of EEGLAB proposes three measures success for the components produced by 23 algorithms [32] using real EEG data. The first of which is a measure of independence based on the log-Likelihood function:

$$LL = -[\ln(|\det(W)|) - \sum\sum \frac{\cosh(S)}{N} - M.\ln(\pi))]$$

Where, W is the mixing matrix, S, the estimated components, N the number of data points and M the number of components. This however isn't a true measure of independence but more a measure of similarity to $\frac{1}{\cosh(S)}$ which is the assumed source activity probability distribution. The second measure used was a percentage measure of returned components having equal dipole projection to the scalp. The third measure of success used was the Amari measure [7] in which success is given by;

$$E_1 = \sum_{i=1}^{n}(\sum_{j=1}^{n}\frac{|p_{ij}|}{max_k|p_{ik}|} - 1) + \sum_{j=1}^{n}(\sum_{i=1}^{n}\frac{|p_{ij}|}{max_k|p_{kj}|} - 1)$$

Where $P = p_{ij} = WA$

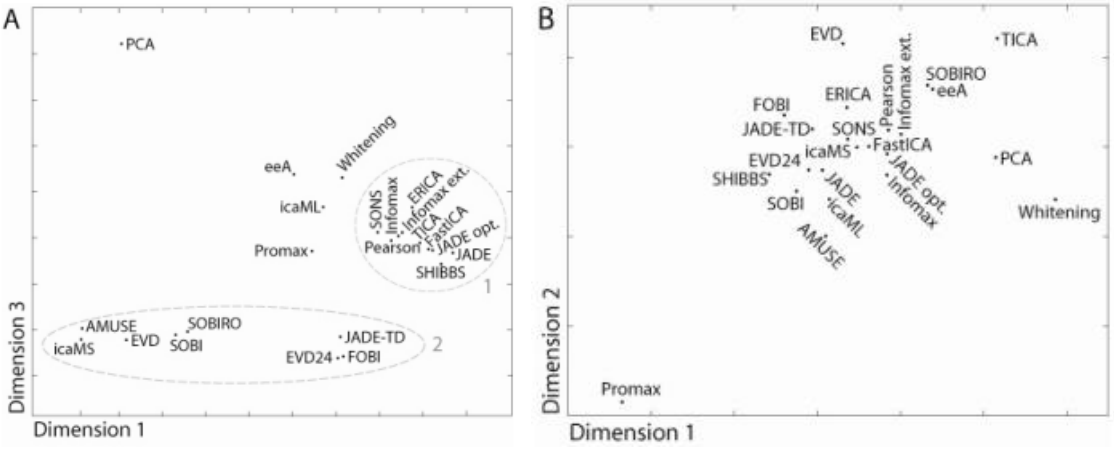


Figure 5) Comparison of all 23 decompositions produced by the algorithms used by Delorme et al, where the results are projected onto axis of maximal interest and their distances calculated using the Amari measure. 5a) group 1 are instantaneous ICA (ICA through optimisation) and produced many near dipole components. Group 2 are time domain methods of ICA. [32]

Many authors use this Amari measure of distance between estimated components, but over many different datasets such as adult ECG [33], foetal ECG [34], and EEG [35], [36].

A commonly used approach in the field of audio separation is to calculate the cross-correlation of the extracted signal with the expected response [37], [38]. Hence;

$$Cross - Corr(x,y) = \frac{Cov(x,y)}{\sqrt{Cov(x,x)Cov(y,y)}}$$

However due to limitations in the literature it is difficult to create a full picture of what algorithms to implement in audio situations as [37], [38] show only that FastICA and the infomax approach (also known as runICA) perform poorly.

Another common approach to measuring the success of an ICA algorithm is to calculate the Root mean squared error between the estimation of the component and the actual component [39] [40]. Hence:

$$RMSE = \sqrt{\frac{1}{k}\sum_{i}^{k}[x(i) - w^{-1}y(i)\,]^2}$$

This measure is effective at comparing the error between the actual and estimated components but requires knowledge of the actual components to calculable. The author uses a rearrangement of this formula to create a similar measure of the success of an algorithm at producing independent components.

Results

To test the effectiveness of solutions produced by the various independent component algorithms, a consistent test was required, one that was simplistic yet effective. This report proposes the use of the sum of squared errors across all components, hence;

$$SSE(\boldsymbol{X}) = \sum_{i}^{m}\sum_{j}^{n}[x_i(j)^2 - \{w_i^{-1}y(j)\}^2]$$

This measure is effective as it is easily normalised between 0 and 1 and averaged across the time series to produce an average error per time point. With this test, 6 algorithms were tested; SOBI, Efficient FastICA (EFICA), Equivalent robust ICA (ERICA), TDSEP, ADAPTICA (a H-J method) and JADE, all of which have been discussed earlier. Each algorithm was implemented in MATLAB and were tested on 4 different datasets:

- ASIN10 contains 10 1000 samples long sinusoids increasing in frequency. This dataset is chosen as it has a high temporal correlation across all components and thus ICA algorithms that assume strict independence may have difficulty acquiring accurate independent components.
- NBAND5 contains 5 radio frequency samples of 2000 samples. These radio frequencies are aimed at mimicking the cocktail party problem and are considered easy for an independent component analysis algorithm to solve.
- ABIO7 contains 7 biological signals, such as, breathing, heartbeat and EEG. This dataset was chosen as the information has little to no temporal dependencies and has solely independent information, thus should be difficult for algorithms that assume temporal correlation but simple for those that assume “strict independence” across components.
- EEG19 contains 17 EEG channels and 2 EMG channels. Where appropriate channels are constantly subject to muscle and ocular artefacts. This is a test of the applicability of the algorithms to real world situations.

	ASIN10	NBAND5	ABIO7	EEG19
SOBI	3.9E-05	8.9E-09	0.0003	0.0056
EFICA	0.00688	0.0005	5E-05	0.0065
ERICA	0.00628	0.00095	0.0008	0.0044

TDSEP	3.7E-24	1.1E-08	0.0003	0.0056
ADAPTICA	0.00621	0.00146	0.0006	0.0054
JADE	0.004	0.00041	1E-04	0.0063

Table 10) shows the error per point in time for each of the algorithms for each dataset tested

From the data, there is a significant difference between the estimations produced by each algorithm for the different datasets and this will be discussed further in the discussion.

Discussion

David Wolpert's no free lunch theorem states that there is no perfect algorithm for any problem, instead there are compromises and a plane of possible optimisations for a given problem [41]. This is apparent in the ICA problem as no single algorithm outperforms all the others, there are always compromises as assumptions on the data are modified and methods of optimisation change across algorithm. This is an absolute reflection of the issue of ICA application, therefore it is paramount that the user understands the nature of the solution provided by an ICA algorithm. From table 1. Methods that rely on "hard" measures of independence provide good approximations for the ABIO7 dataset however in systems with temporal dependencies they provide poor approximations. This is because the methods that assume strict independence across components don't use any information about the temporal dynamics within the input data, therefore in cases where there are no temporal dynamics they achieve a better approximation than those that do rely on temporal dynamics. Although not directly tested in this paper it is known that methods of joint diagonalization tend to be faster than methods that rely on strict optimization, though this is only in cases of bulk Independent component analysis []. Bulk, meaning that a large amount of data regarding the component interactions is required, for example, in the case of people talking, many time samples of people talking is required for bulk algorithms to find accurate solutions. Many bulk algorithms see significant increase in approximation error when not supplied sufficient samples of the input data [].

A new area of ICA research is that of recursive ICA algorithms, originally proposed by J. Cardoso (creator of JADE) in his algorithm EASI [] they feature a move away from the bulk processing approach to ICA, instead iteratively approximating the unmixing matrix over a series of time steps. This allows for approximations of non-stationary sources as well as an increase in the temporal efficiency of ICA algorithms at the cost of estimate accuracy [].

Conclusion

Project Commentary

The project developed steadily throughout the year, with its aims changing as new information became available. The project, in its final form is similar to the one initially described in the PID, however distinctly different. Originally the project was aimed at developing novel methods of Recursive online independent component analysis, however as research continued it became

increasingly clear that in order to understand how recursive online methods of ICA worked it would be important to understand how all ICA methods worked and what made them unique and applicable in different situations. Hence it became important to test and verify conclusions, citing relevant papers, and understand the current developments in order to provide a stable foundation for future work in the field.

Social, Legal and Ethical Issues

Intellectual Property Laws with respect to algorithms must be considered whenever an individual works with patented software. All algorithms used in this project for example allow use in academic environments (provided they are cited) but not in commercial by the request of the authors. The problem of intellectual property ownership in algorithms is simple but has deep implications, for example; What does it mean to own an algorithm? Can you own mathematics, or a thought process? How similar does an algorithm must be before it is considered a breach of IP law? The law ultimately decides, however the lines describing the bounds of IP law are intentionally vague with algorithms being seen as complex assets and receiving varying degrees of legal protection.

In the wider sense, Independent component analysis can have both beneficial and deeply unsettling implications on society. Where ICA can reduce high-dimensional noisy data into independent signals, it is highly beneficial in the field of acoustic signal processing. This, comes as a double edges sword however, as it both allows individuals greater freedom using hand-free devices whilst it also enables mass communication surveillance through embedded microphones in the environment. This mass surveillance opens questions as to whether it is acceptable for governments to listen to their populous, either actively or passively? That is, if they had the computational power. As this is not a philosophy of ethics report, full analysis of these issues is out of the scope of this report, however, it is important to note the importance of recognising and understanding the potential implications novel technologies and understandings can have on wider society.

Reflection

Given the open-ended nature of the project, I found that the freedom in direction gave me the possibility to explore novel directions, developing understanding, and experimenting with the field and its possibilities. This kind of developing project is not something that could be planned and prepared, as many of the prerequisite knowledge required to understand the topic of interest were not common knowledge to the author, and thus an agile approach to the project was required. Though there were set backs through the project, none of them were significant enough not to be redeveloped around and thus risk mitigation was achieved effectively throughout the project.

References

[1] A. Hyvaarinen and E. Oja, “Independent Component Analysis: algorithms and applications,” Helsinki University of Techology.

[2] Y.-M. Chen and Y.-C. Wu, “Modified recursive least-squares algorithm for parameter identification,” *International Journel of Systems Science,* vol. 23, no. 2, pp. 187-205, 1992.

[3] Y. Wang, C. Li and C. Tian, “Modified recursive least squares algorithm for sparse system identification,” in *7th International Conference on Modelling, Identification and Control (ICMIC)*, 2015.

[4] A. Muddasir and M. Aqil, “QR decomposition based recursive least square adaptation of autoregressive EEG features,” in *International Conference on Intelligent Systems Engineering (ICISE)*, 2016.

[5] C. Jutten and J. Herault, “Blind Separation of Sources, Part I: Adaptive Algorithm based on Neuromimetic Architecture,” *Signal Processing,* pp. 1-10, 1991.

[6] A. Cichocki, R. E. Bogner, L. Moszczyń ski and K. Pope, “Modified Herault-Jutten Algorithms for Blind Source Separation,” *Digital Signal Processing,* vol. 7, pp. 80-93, 1997.

[7] S. Amari, A. Cichocki and H. H. Yang, “A New Learing Algorithm For Blind Source Separation,” in *Advances in Neural Information Processing Systems*, MIT Press, 1996, pp. 757-763.

[8] A. J. Bell and T. J. Sejnowski, “An Information-Maximisation Approach To Blind Source Separation and Blind Deconvolution,” *Neurocomputation,* pp. 1129-1159, 1995.

[9] M. Kendall and A. Stuart, “The Advanced Theory of Statistics,” Charles Griffin & Company, 1958.

[10] S. Cruces, L. Castedo and A. Cichocki, “Robust blind source separation algorithms using cumulants,” *Neurocomputing,* vol. 49, pp. 87-118, 2002.

[11] A. Hyvarinen, “Fast and Robust Fixed-Point Algorithms for Independent Component Analysis,” *IEEE Transactions on Neural Networks,* vol. 10, no. 3, pp. 626-634, 1999.

[12] P. Tichavsky, Z. Koldovsky and E. Oja, “Performance Analysis of the FastICA Algorithm and Cramer-Rao Bounds for Linear Independent Component Analysis,” *IEEE Transactions of Signal Processing,* vol. 54, no. 4, pp. 1189-1203, 2006.

[13] V. Zarzoso and P. Comon, “Comparative Speed Analysis of FastICA,” in *International Conference on Independent Component Analysis and Signal Separation*, 2007.

[14] C. Fan and H. J. Baoqiang Wang, “A New FastICA Algorithm with Symmertic Orthogonalization,” in *IEEE International Conference on Communications, Circuits and Systems*, 2006.

[15] D. Zhang and H. Shao, “Improved FastICA Algorithm based on Symmetric Orthogonalization,” in *IEEE 11th International Conference on Signal Processing*, 2012.

[16] J. H. Friedman and J. W. Tukey, “A Project Pursuit Algorithm for Exploratory Data Analysis,” *IEEE Transactions on Computers,* Vols. C-23, no. 9, pp. 881-890, 1973.

[17] F. Moreto and M. Ramirez, “A Projection Pursuit Apporach to Blind Source Separation,” in *Proceedings of the International Workshop of Telecommunications*.

[18] V. Khorani, N. Forouzideh and A. Nasrabadi, “Artificial neural network weights optimization using ICA, GA, ICA-GA and R-ICA-GA: Comparing performances,” in *IEEE Workshop on Hybrid Intelligent Models and Applications*, 2011.

[19] A. Hyvarinen, “New Approximations of Differential Entropy for Independent Component Analysis and Projection Pursuit,” *Advances in Neural Information Processing Systems,* vol. 10, pp. 273-279, 1998.

[20] S. Cruces-Alvarez, A. Cichocki and L. Castedo-Ribas, “An Iterative Inversion Approach to Blidn Source Separation,” *IEEE Transactions on Neural Networks ,* vol. 11, no. 6, pp. 1423-1437, 2000.

[21] R. Andre, T. Trainini, X. Luciani and E. Moreau, “A Fast Algorithm for Joint Eigenvalue Decomposition of Real Matrices,” in *23rd European Signal Processing Conference* , 2015.

[22] W.-J. Liu, D.-Z. Feng and W.-K. Nie, “A Jacobi-like joint diagonalization method by one-dimensional optimization,” *Signal Processing,* vol. 119, pp. 174-184, 2016.

[23] A. Bigan and K. S. Perinkulam, “Some Gradient Based Joint Diagonalization Methods for ICA,” in *International Conference on Independent Component Analysis and Signal Separation*, 2004.

[24] L. Tong, V. Soon, Y. Huang and R. Liu, “AMUSE: a new blind identification algorithm,” in *IEEE International Symposium on Circuits and Systems*, 1990.

[25] A. Ziehe and K.-R. Muller, “TDSEP - An Efficient Algorithm For Blind Separation Using Time Structures,” in *Perspectives in Neural Computing*, 1998, pp. 675-680.

[26] A. Belouchrani, K. Abed-Meraim, J. F. Cardoso and E. Moulines, “A Blind Source Separation Technique Using Second-Order Statistics,” *IEEE Transactions on Signal Porcessing,* vol. 45, no. 2, pp. 434-444, 1997.

[27] J. F. Cardoso and A. Souloumiac, “Blind beamforming for non-Gaussian signals,” *IEE proceedings on Radar and Signal Processing,* vol. 140, no. 6, pp. 362-369, 1993.

[28] Z. Koldovsky and P. Tichavsky, “Methods of Fair Comparison of Performance of Linear ICA Techniques in Presence of Additive Noise,” in *IEEE International Conference on Acoustics Speech and Signal Processing Proceedings*, 2006.

[29] Z. Koldovský and P. Tichavsky, “A Comparison of Independent Component and Independent Subspace Analysis Algorithms,” in *17th European Signal Processing Conference*, Glasglow, Scotland, 2009.

[30] D. T. Pham and J. F. Cardoso, “Blind separation of instantenous mixtures of nons tationary sources,” *IEEE Transactions on Signal Processing,* vol. 49, no. 9, pp. 1837-1848, 2001.

[31] P. Tichavsky, A. Yeredor and Z. Koldovsky, “A Fast Asymptotically Efficient Algorithm for Blind Source Separation of a Linear Mixture of Block-Wise Stationary Autoregressive Processes,” in *IEEE International Conference on Acoustics, Speech and Signal Processing*, 2009.

[32] A. Delorme, J. Palmer, R. Oostenveld and S. Makeig, “Comparing Results of Algorithms Implementing Blind Source Separation of EEG Data,” A. Delorme, 2007.

[33] H. Kasturiwale and Z. Mizwan, “Comparison and Performance Analysis of various ICA Algorithms for ECG Signals,” *International Journal of Engineering Research and Technology,* vol. 3, no. 4, pp. 674-677, 2014.

[34] S. D. Parmer and B. Unhelkar, “A Comparison of the Perofrmance of ICA Algorithms for Fetal ECG Extraction using Time Domain Multiple-Sources Interference Data,” *International Journal of Recent Trends in Engineering and Technology,* vol. 3, no. 1, pp. 11-13, 2010.

[35] X. Giannakopoulos, J. Karhunen and E. Oja, “An Experimental Comparison of Neural ICA Algorithms,” in *ICANN 98: Proceedings of the 8th International Conference on Artificial Neural Networks*, Skovde, Sweden, 1998.

[36] J. Walters-Williams and Y. Li, “Performance Comparison of Known ICA Algorithms to a Wavelet-ICA Merger,” *International Journal of Signal Processing,* vol. 5, no. 3, pp. 80-92, 2011.

[37] K. Mohanaprasad and P. Arulmozhivarman, “Comparison of FastICA and Gradient Algorithms of Independent Component Analysis for Separation of Speech Signals,” *International Journal of Engineering and Technology,* vol. 5, no. 4, pp. 3196-3202, 2013.

[38] P. Gorski, “Performance comparison of ICA algorithms for audio blind source separation,” West Pomeranian University of Technology, 2015.

[39] G. R. Naik, D. Kumar and H. Weghon, “Performance Comparison of ICA Algorithms for isometric Hand Gesture Identification using Surface EMG,” in *3rd International Conference on Intelligent Sensors, Sensor Networks and Information*, 2007.

[40] V. Matic, W. Deburchgraeve and S. Van Huffel, “Comparison of ICA Algorithms for ECG Artifcat Removal from EEG Signals,” in *Annual Symposium of the IEEE Photonics Benelux Chapter*, 2009.

[41] D. H. Wolpert and W. G. Macready, “No free lunch theorems for optimization,” *IEEE Transactions on Evolutionary Computation,* vol. 1, no. 1, pp. 67-82, 1997.

[42] P. Tichavsky, Z. Koldovsky, E. Doron, A. Yeredor and G. Gomez-Herrero, “Blind Source Separation By Combining Two ICA Algorithms: Hos-based EFICA and Tiem Structure-based WASOBI,” in *14th European Signal Processing Conference*, Florence, Italy, 2006.

Media References

[M1] http://www.pathpartnertech.com/wp-content/themes/PP/assets/img/articles/Signal-un-mixing-using-BSS.png

[M2] https://en.wikipedia.org/wiki/Skewness#/media/File:Negative_and_positive_skew_diagrams_(English).svg

[M3] http://math.illinoisstate.edu/day/courses/old/312/session0611.gif

[M4] https://www.gerfficient.com/wp-content/uploads/2016/07/pca-vs-ica.png

Appendices

Appendix 1. Gram-Schmidt Orthogonalization

The Gram-Schmidt process is a method for transforming an arbitrary vector $U = \{u_1, \cdots, u_N\}$ in the inner product space D into an orthogonal basis vector $V = \{v_1, \cdots v_n\}$. The method is as follows;

Step 1. Let $v_1 = u_1$

Step 2. $v_2 = u_2 - \frac{<u_2, v_1>}{||v_1||^2} v_1$

Step 3. $v_3 = u_3 - \frac{<u_3, v_1>}{||v_1||^2} v_1 - \frac{<u_3, v_2>}{||v_2||^2} v_2$

Continuing for all of n.

Appendix 2. Cross-Correlation, Convolution and Covariance

Correlation is a measure of interdependence between two variables X and Y. If the two variables are dependent on one another then the measure will be either positive or negative depending on the relationship between. If the measure is zero, then there is no relationship between X and Y. If $X \equiv Y$ then the Cross-Correlation is called the Autocorrelation.

$$X * Y = \int_{-\infty}^{\infty} X^*(t) Y(t + \tau) dt$$

The Correlation Coefficient is another measure, closely related to the correlation which is used to measure the interdependence of two variables.

$$CorrCoef(X, Y) = \frac{E[(X_i - \bar{X})(Y_i - \bar{Y})]}{\sigma_X \sigma_Y} \equiv \frac{Cov(X, Y)}{\sigma_X \sigma_Y}$$

Appendix 3. Singular Value Decomposition

Singular Value Decomposition is a method of dimensional reduction, creating 2 rotational matrices and a vector of eigenvalues from a single matrix. For a matrix $Y = M \times N$ this can be shown mathematically as $SVD(y) = USV^T$, the method to which U, S and V can be found is described below;

1) Find the N non-zero eigenvalues, λ_i of the matrix $C = y^T y$ and construct the diagonal matrix, S out of the square roots of the eigenvalues in descending order of magnitude.
2) Find the orthogonal eigenvectors of the covariance matrix C corresponding to the obtained eigenvalues and arrange them in the same order. This order of eigenvectors becomes the matrix V.
3) Find the remaining $M - N$ column vectors of the matrix U through $u_i = s_i{}^{-1} X v_i$ where $i = 1{:}N$.
4) Add the rest of the columns of U using Gram-Schmidt orthogonalization (see Appendix 3.)

Appendix 4. Principle Component Analysis

Principle Component Analysis hereby known as PCA is a transform from the data domain to the covariance domain, typically plotted as the spread of the eigenvalues. The aim is to find the x largest eigenvalues that represent the majority of the variance in the data. The transform follows a simple method which will be shown in a 2-dimensional case below;

- Subtract the mean from the data set A, centralizing the data set around the (0,0) point.
- Calculate the Covariance of the new dataset A via;

$$Cov(x, y) = \frac{(x-\mu_x)(y-\mu_y)}{\sigma_x \sigma_y}$$

- Find the eigenvalues and eigenvectors of the dataset A and arrange the eigenvalues in order of magnitude.
- Transpose the vector and multiply it by the original data set A in order to see the spread of the data.

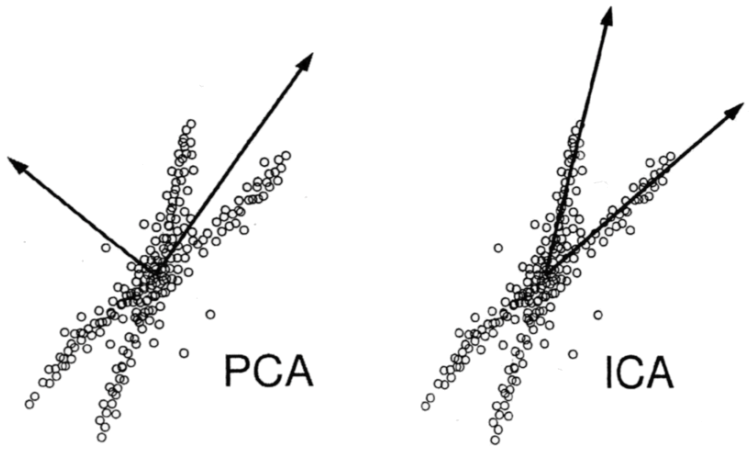


Figure 6. [M4]